



Problem solving with factors, multiples, squares and cubes

Task A: Solve problems with squares and cubes

- 1) a) What is the largest square you can make with 80 blocks?

$8 \times 8 = 64$, 81 would be needed for 9

- b) What is the largest square you can make with 200 blocks?

$14 \times 14 = 196$, 225 would be needed for 15

- c) What is the largest cube you can make with 200 blocks?

$5 \times 5 \times 5 = 125$, 216 would be needed for 6

- d) What is the largest cube you can make with 350 blocks?

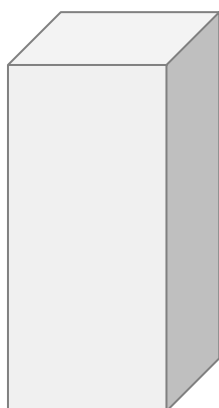
$7 \times 7 \times 7 = 343$, 512 would be needed for 8

- e) Justify why 8 is not a square number.

There are not two repeated integers whose product is 8, $2 \times 2 = 4$ and $3 \times 3 = 9$

- 2) The volume of a cuboid is 90 cm^3 .

If the dimensions are all integers, what could they be?



$1 \times 1 \times 90$
 $1 \times 2 \times 45$
 $1 \times 3 \times 30$
 $1 \times 5 \times 18$
 $1 \times 6 \times 15$
 $1 \times 9 \times 10$

$2 \times 9 \times 5$
 $6 \times 3 \times 5$
 $2 \times 3 \times 15$
 $10 \times 3 \times 3$



Task B: Solve problems with factors and multiples

- 1) Each of the following rows contains consecutive numbers which satisfy the condition given. No number is greater than 20.

a)

9 three factors	10 sum of digits is 1	11 odd	12 multiple of 3
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b)

14 sum of digits is 5	15 sum of digits is 6	16 five factors	17 prime
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c)

6 four factors	7 prime	8 four factors	9 three factors
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- 2) Use the product of prime factors to complete these grids.

a)

2	7	11	154
2	5	3	30
5	2	2	20
20	70	66	

$$154 = 2 \times 7 \times 11$$

$$30 = 2 \times 3 \times 5$$

$$20 = 2 \times 2 \times 5$$

$$70 = 2 \times 5 \times 7$$

$$66 = 2 \times 3 \times 11$$

- b) Use your calculator to write these numbers as a product of prime factors to complete the grid.

5	5	11	275
3	11	3	99
7	2	5	70
105	110	165	

